



Communications and Networks

Passband Transmission

Part II: Detection and Error Analysis

Sheng Zhou



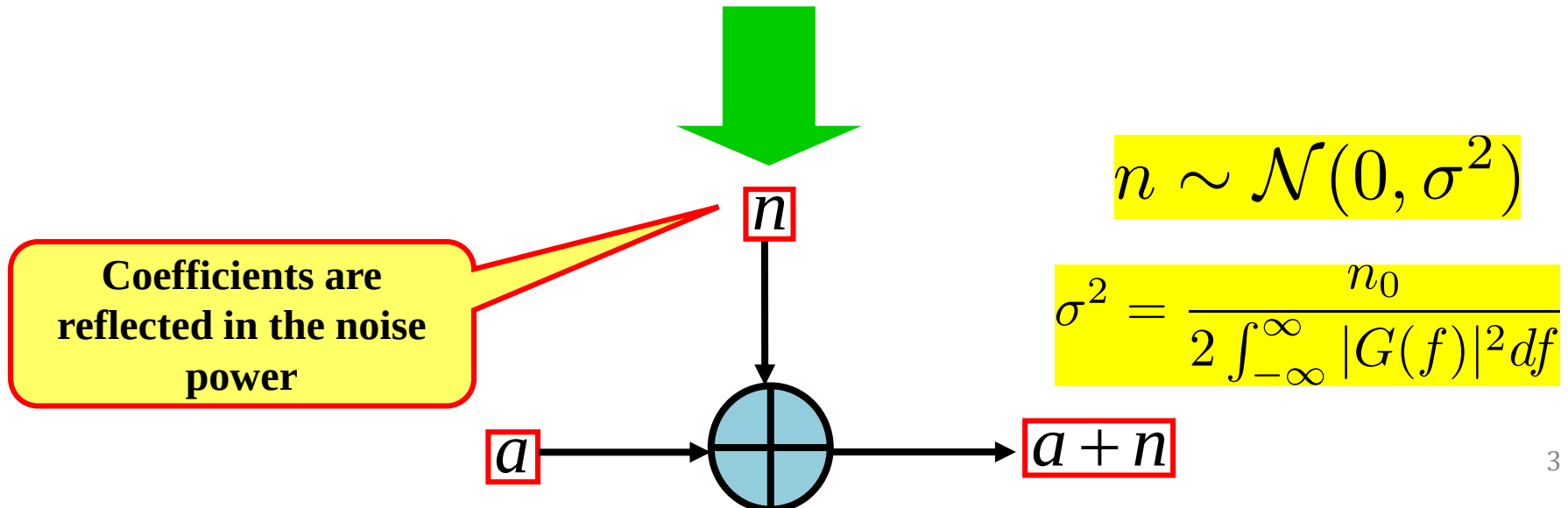
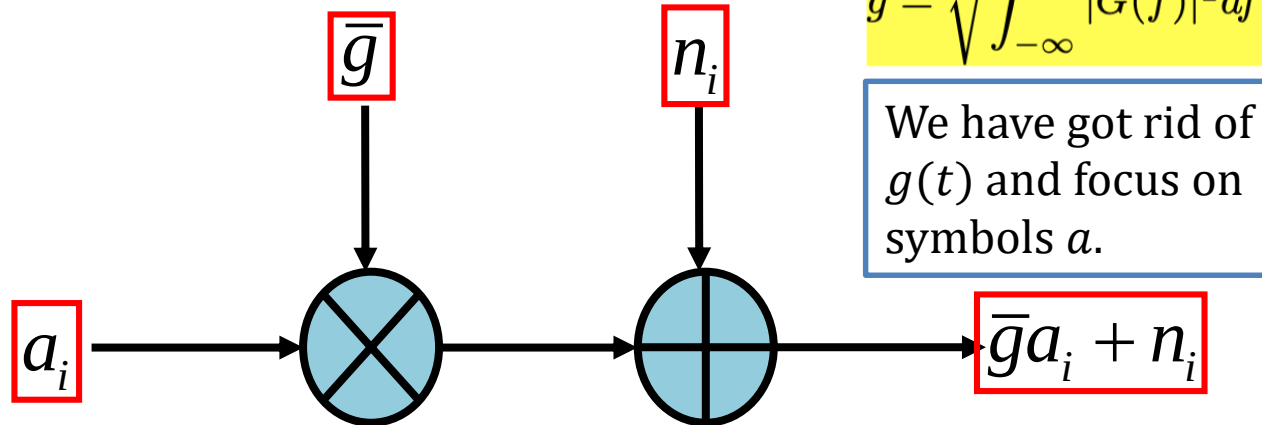
Outline

- Baseband equivalent AWGN channel
- Detection and error analysis
 - ▶ MASK
 - ▶ MPSK
 - ▶ MQAM
 - ▶ MFSK



Noisy Channel

Recall Lecture 5





Passband AWGN Channel

- Question: What is the baseband representation of the original passband AWGN channel?
- Passband $y(t) = x(t) + n(t)$
- **Complex** baseband symbol model $\mathbf{y} = \mathbf{a} + \mathbf{n}$
- Let passband $n(t)$ has a PSD $\frac{n_0}{2}$
- Then the complex baseband $\mathbf{n}$ has variance n_0
 - ▶ **Each dimension** (inphase/quadrature) has variance $\frac{n_0}{2}$
 - ▶ The two dimensions are uncorrelated with each other



Passband AWGN Channel

■ Ready for the analysis in next section!

■ For 1D modulation (M-ASK, M-FSK)

$$c = \int_{-\infty}^{+\infty} g^2(t) dt$$

▶ $E_s = c\mathbb{E}(|\mathbf{a}|^2)$

▶ $\sigma^2 = \frac{n_0}{2}$

$$\frac{E_s}{n_0} = \frac{\overline{a_n^2}}{2\sigma^2}$$

Same as
baseband

$$\frac{\overline{a_n^2}}{\sigma^2} = \frac{S}{N} = \frac{2E_s}{n_0}$$

■ For 2D modulation (M-PSK, M-QAM): **Different!**

▶ $E_s = c\mathbb{E}(|\mathbf{a}|^2) = c\mathbb{E}(|a_I|^2 + |a_Q|^2)$

▶ $\sigma^2 = n_0 = 2\sigma_I^2 = 2\sigma_Q^2, \sigma_I^2 = \sigma_Q^2 = \frac{n_0}{2}$

$$\frac{E_s}{n_0} = \frac{\overline{(a_n^2 + b_n^2)}}{\sigma^2}$$

$$\frac{E_s}{n_0/2} = \frac{\overline{(a_n^2 + b_n^2)}}{\sigma_I^2} = \frac{\overline{(a_n^2 + b_n^2)}}{\sigma_Q^2}$$



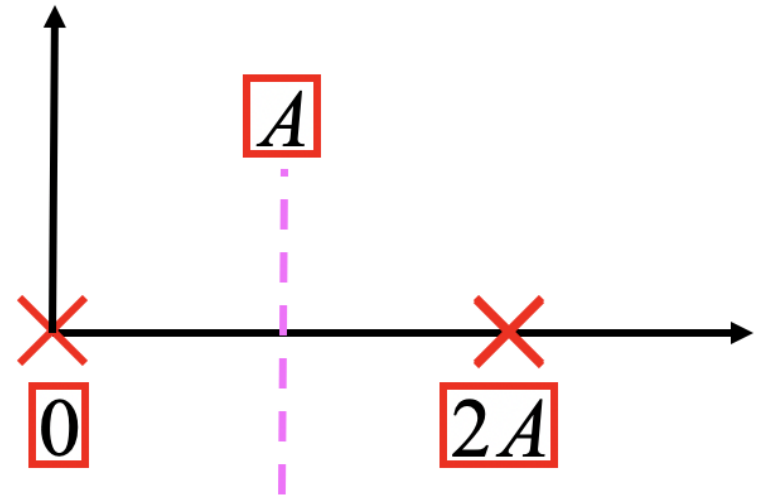
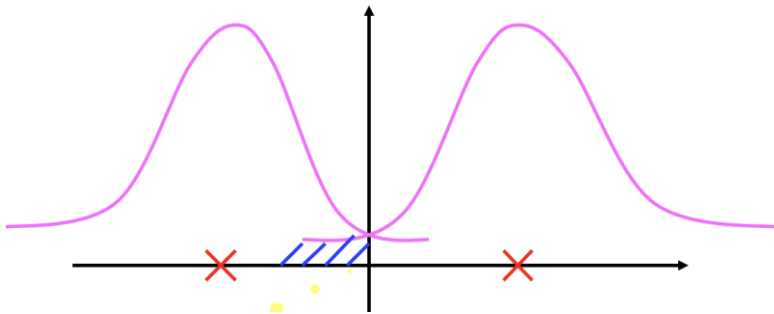
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Binary Signal Detection

- In Lecture 5, we discussed binary antipodal and non-antipodal signal detection.





Review the Steps of Calculating SEP

- 1. Draw the symbols
 - ▶ With inter-symbol distance $2A$
- 2. Decide the detection thresholds
 - ▶ Distance between the symbol to its thresholds is A
- 3. Derive the conditional error probability for each symbol **as the function of A/σ**
 - ▶ **Case by Case**
- 4. Use A to characterize E_s
 - ▶ Obtain **A/σ** as the function of **E_s/n_0**
Choose correct σ !! (from which dimension?)
- 5. Get the SEP as a function of E_s/n_0

$$\begin{array}{ll} \text{1-D} & \frac{E_s}{n_0} = \frac{\overline{a_n^2}}{2\sigma^2} \\ \text{2-D} & \frac{E_s}{n_0} = \frac{\overline{(a_n^2 + b_n^2)}}{\sigma^2} \\ & \frac{E_s}{n_0/2} = \frac{\overline{(a_n^2 + b_n^2)}}{\sigma_I^2} \end{array}$$



Outline

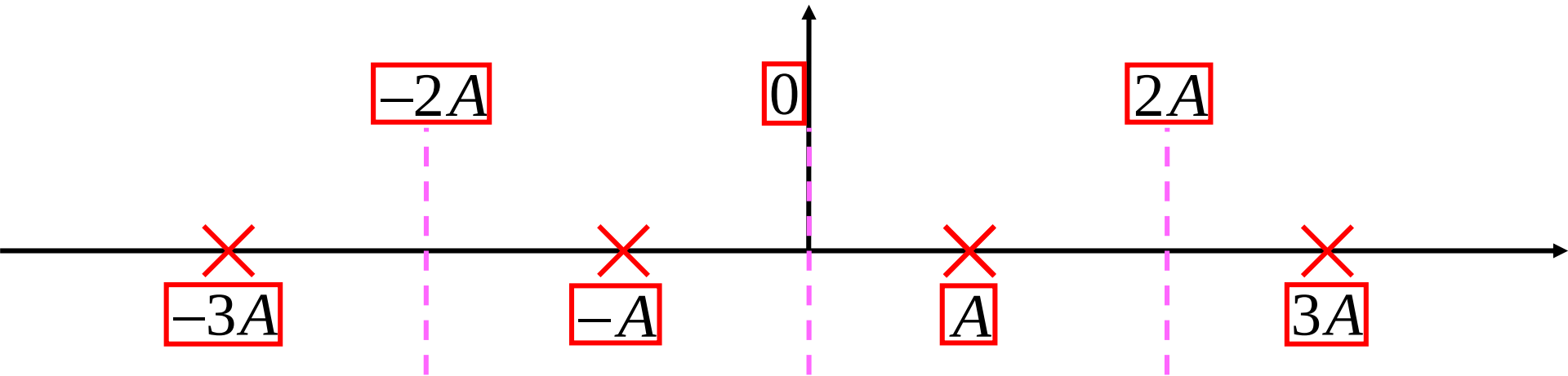
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M -ary Antipodal Signal (*Even* M)

- 1. **Even** number of symbol elements, $|A| = M = 2k$
 $\{\pm A, \pm 3A, \dots, \pm(M-1)A\}$
- 2. Thresholds are
 $\{0, \pm 2A, \dots, \pm(M-2)A\}$

Minimum distance rule!



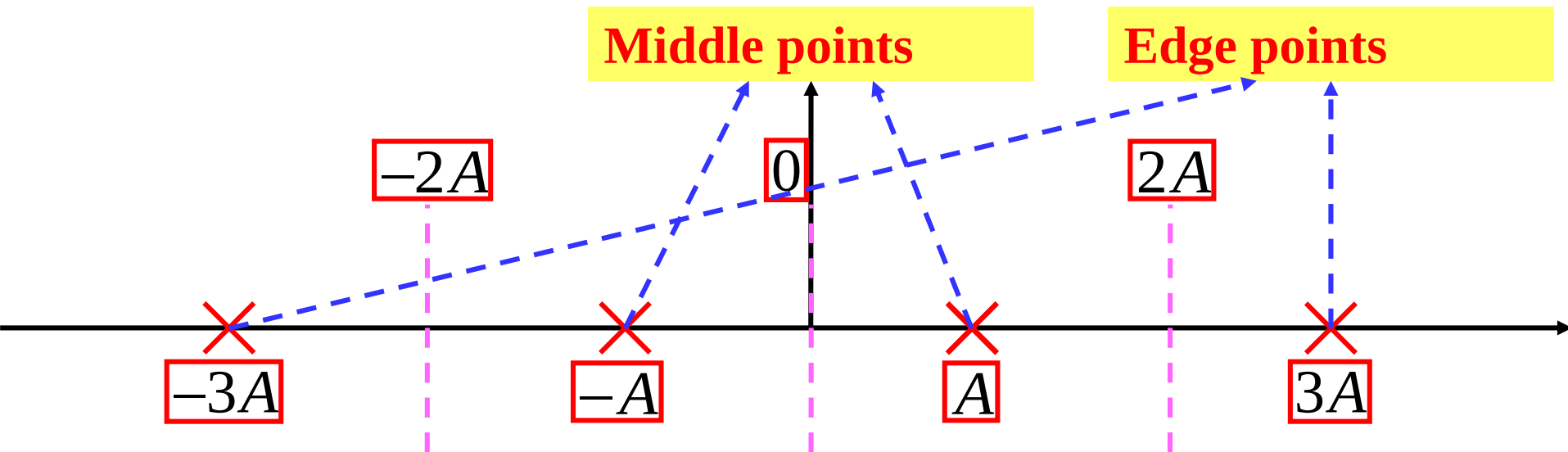
Constellation for baseband PAM and passband ASK



M -ary Antipodal Signal (*Even* M)

3. Derive the conditional error probability

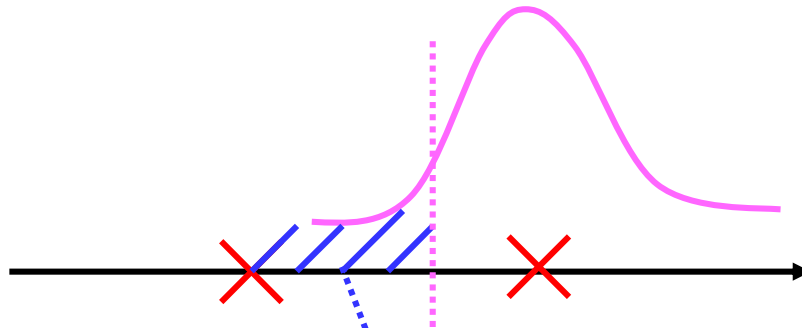
► Note: when $M > 2$, there are two cases!





M -ary Antipodal Signal (*Even* M)

- For the two edge points
 - ▶ Error can only occur on one side



- ▶ Conditional error

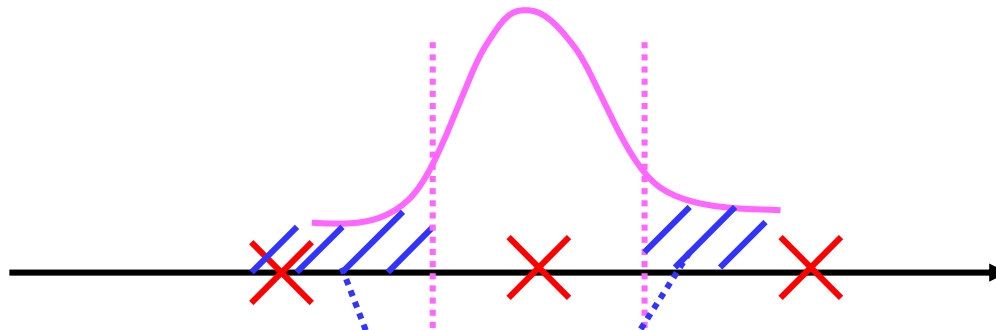
$$Q\left(\frac{A}{\sigma}\right)$$



M -ary **Antipodal** Signal (*Even* M)

■ For the middle points

► Error can only occur on both sides of the symbol



► Conditional error

$$2Q\left(\frac{A}{\sigma}\right)$$



M -ary **Antipodal** Signal (*Even* M)

■ 3. Average error probability for all symbols

- ▶ 2 edge points each with conditional error probability

$$Q\left(\frac{A}{\sigma}\right)$$

- ▶ $M-2$ middle points each with conditional error probability

$$2Q\left(\frac{A}{\sigma}\right)$$

- ▶ **Remember equal prob. Assumption! Then the average SEP**

$$P_s = \frac{1}{M} \left[2Q\left(\frac{A}{\sigma}\right) + (M-2)2Q\left(\frac{A}{\sigma}\right) \right] = \frac{2(M-1)}{M} Q\left(\frac{A}{\sigma}\right)$$



M -ary **Antipodal** Signal (*Even* M)

■ 4. Use A to characterize E_s

$$\begin{aligned} E_s &= \frac{2}{M} \left[(A)^2 + (3A)^2 + \dots + ((M-1)A)^2 \right] \\ &= \frac{2A^2}{M} [1^2 + 3^2 + \dots + (M-1)^2] \\ &= \frac{2A^2}{M} \frac{M}{6} (M^2 - 1) \\ &= \frac{M^2 - 1}{3} A^2 \end{aligned}$$



M -ary Antipodal Signal (*Even* M)

- 5. Obtain A/σ as the function of E_s/n_0

$$\frac{E_s}{n_0} = \frac{\frac{M^2 - 1}{3} A^2}{2\sigma^2} \longleftrightarrow \frac{A}{\sigma} = \sqrt{\frac{6E_s}{(M^2 - 1)n_0}}$$

- Use the result from step 3:

$$P_s = \frac{2(M - 1)}{M} Q\left(\frac{A}{\sigma}\right)$$

$$P_s = \frac{2(M - 1)}{M} Q\left(\sqrt{\frac{6E_s}{(M^2 - 1)n_0}}\right)$$

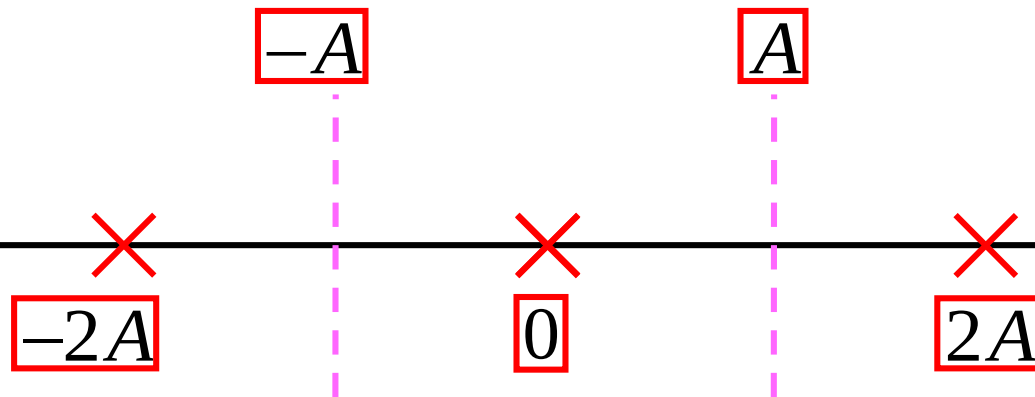
What if M is odd?



M -ary **Antipodal** Signal (*Odd* M)

- ▶ 1. **Odd** number of symbol elements
 $\{0, \pm 2A, \pm 4A, \dots, \pm(M-1)A\}$
- ▶ 2. Thresholds are

$$\{\pm A, \pm 3A, \dots, \pm(M-2)A\}$$

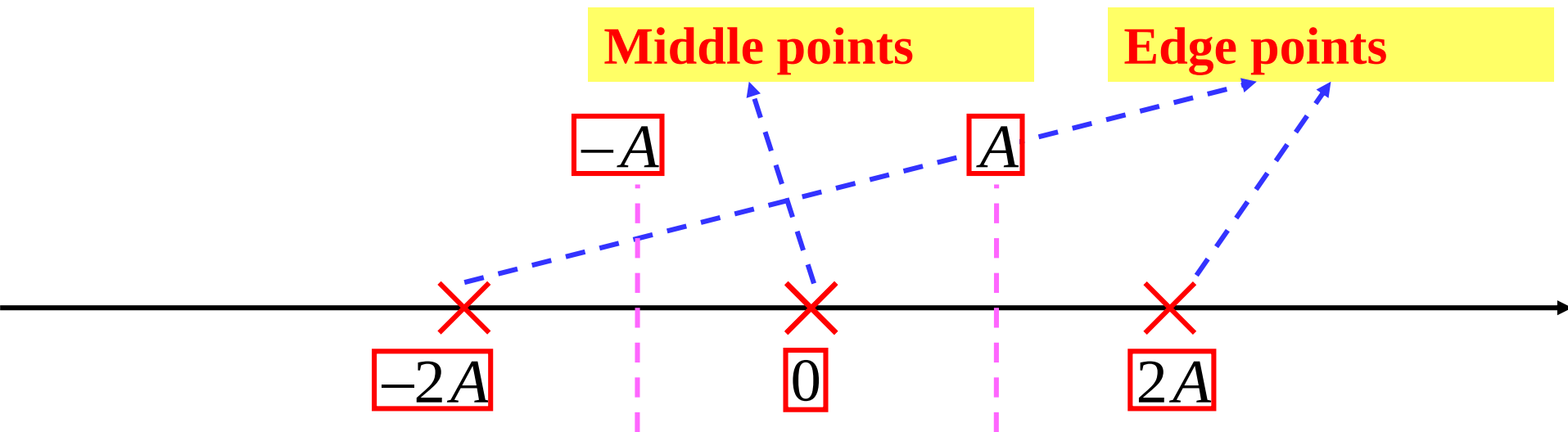




M -ary Antipodal Signal (*Odd* M)

3. Derive the conditional error probability

► Note: **when $M > 2$, there are two cases!**



$$P_s = \frac{1}{M} \left[2Q \left(\frac{A}{\sigma} \right) + (M - 2) 2Q \left(\frac{A}{\sigma} \right) \right] = \frac{2(M - 1)}{M} Q \left(\frac{A}{\sigma} \right)$$



***M*-ary Antipodal Signal (*Odd M*)**

■ 4. Use A to characterize E_s

$$\begin{aligned} E_s &= \frac{1}{M} \left\{ 0 + 2[(2A)^2 + (4A)^2 + \dots + ((M-1)A)^2] \right\} \\ &= \frac{2A^2}{M} [2^2 + 4^2 + 6^2 + \dots + (M-1)^2] \\ &= \frac{2A^2}{M} \times 4 \left[1^2 + 2^2 + 3^2 + \dots + \left(\frac{M-1}{2} \right)^2 \right] \\ &= \frac{8A^2}{M} \frac{1}{6} \frac{M-1}{2} \left(\frac{M-1}{2} + 1 \right) \left(2 \times \frac{M-1}{2} + 1 \right) \\ &= \frac{M^2 - 1}{3} A^2 \end{aligned}$$



M-ary **Antipodal** Signal (*Odd M*)

- 5. Obtain A/σ as the function of E_s/n_0

$$\frac{E_s}{n_0} = \frac{\frac{M^2 - 1}{3} A^2}{2\sigma^2} \longleftrightarrow \frac{A}{\sigma} = \sqrt{\frac{6E_s}{(M^2 - 1)n_0}}$$

- Use the result from step 3:

$$P_s = \frac{2(M - 1)}{M} Q\left(\frac{A}{\sigma}\right)$$

$$P_s = \frac{2(M - 1)}{M} Q\left(\sqrt{\frac{6E_s}{(M^2 - 1)n_0}}\right)$$

The same for Odd and Even M !



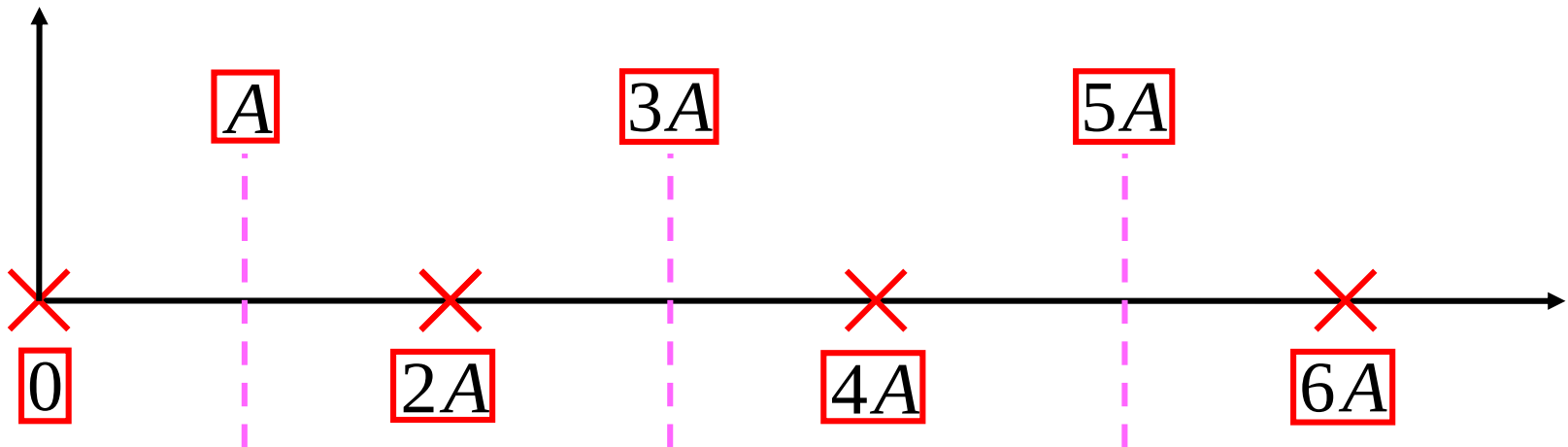
M-ary Non-Antipodal Signal

- 1. Set of symbol elements

$$\{0, 2A, 4A, \dots, 2(M-1)A\}$$

- 2. Thresholds are

$$\{A, 3A, 5A, \dots, (2M-3)A\}$$



- 3.
$$P_s = \frac{1}{M} \left[2Q\left(\frac{A}{\sigma}\right) + (M-2)2Q\left(\frac{A}{\sigma}\right) \right] = \frac{2(M-1)}{M} Q\left(\frac{A}{\sigma}\right)$$



M -ary Non-Antipodal Signal

■ 4. Use A to characterize E_s

$$\begin{aligned} E_s &= \frac{1}{M} \sum_{m=0}^{M-1} (2mA)^2 \\ &= \frac{4A^2}{M} [1^2 + 2^2 + \dots + (M-1)^2] \\ &= \frac{4A^2}{M} \frac{(M-1)M(2M-1)}{6} \\ &= \frac{2(M-1)(2M-1)}{3} A^2 \end{aligned}$$



M-ary Non-Antipodal Signal

- 5. Obtain A/σ as the function of E_s/n_0

$$\frac{E_s}{n_0} = \frac{2(M-1)(2M-1)}{3} \frac{A^2}{2\sigma^2} \longleftrightarrow \frac{A}{\sigma} = \sqrt{\frac{6}{2(M-1)(2M-1)} \frac{E_s}{n_0}}$$

► Use the result from step 3: $P_s = \frac{2(M-1)}{M} Q\left(\frac{A}{\sigma}\right)$

$$P_s = \frac{2(M-1)}{M} Q\left(\sqrt{\frac{6E_s}{2(M-1)(2M-1)n_0}}\right)$$

Comparing with Antipodal?



Antipodal and Non-Antipodal

■ SEP comparison

▶ Antipodal

$$P_s = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{6E_s}{(M^2-1)n_0}} \right)$$

▶ Non-Antipodal

$$P_s = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{6E_s}{2(M-1)(2M-1)n_0}} \right)$$

With large M , to have the same SEP, non-antipodal requires 6dB more energy per symbol. WHY ????



From SEP to Bit Error Probability (BEP)

- People often care about bit error probability (BEP)
- Recall:
 - ▶ One symbol carries $\log_2 M$ bits
 - ▶ The error of ONE symbol *approximately* results in ONE bit error
 - ▶ This is because: (1) **Gray Encoding**; (2) Under normal SNR, if the detection error occurs, the erroneous output is often the adjacent symbol.

$$P_b \approx \frac{P_s}{\log_2 M}$$

**Approximation, only
precise for binary signal**



SER and BER for M -ary Signals w.r.t E_b/n_0

■ Antipodal

$$P_s = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{6 \log_2 M E_b}{M^2 - 1} \frac{E_b}{n_0}} \right)$$

$$E_s = E_b \log_2 M$$

$$P_b \approx \frac{P_s}{\log_2 M}$$

$$P_b = \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{6 \log_2 M E_b}{M^2 - 1} \frac{E_b}{n_0}} \right)$$

■ Non-antipodal

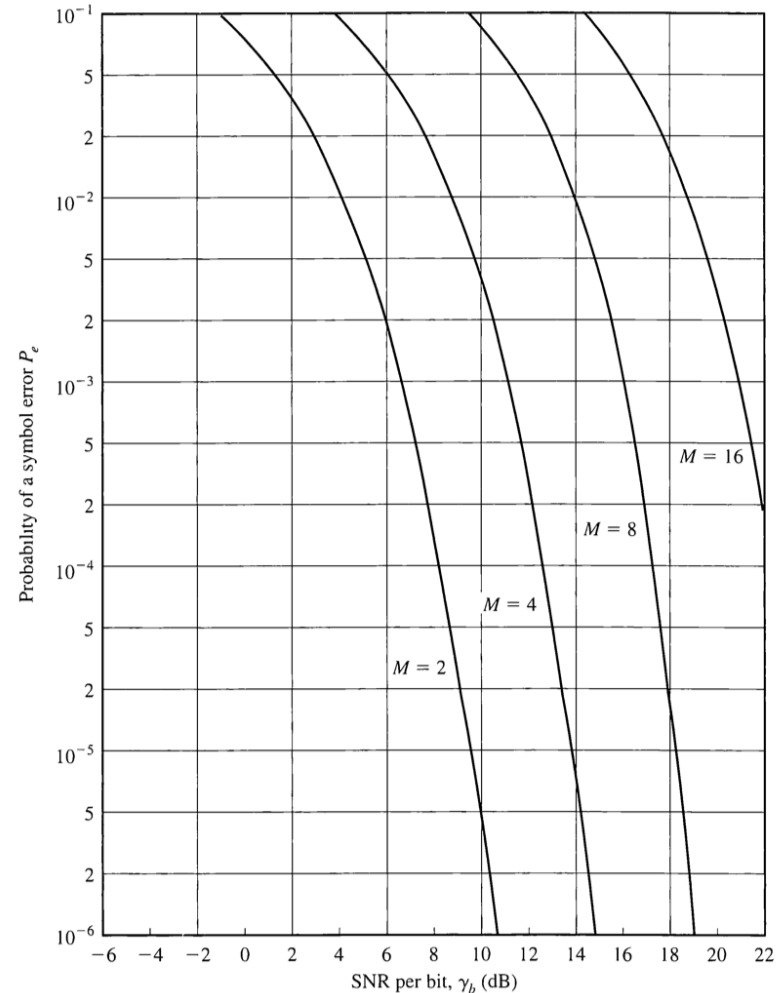
$$P_s = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{3 \log_2 M}{(M-1)(2M-1)} \frac{E_b}{n_0}} \right)$$

$$P_b = \frac{2(M-1)}{M \log_2 M} Q \left(\sqrt{\frac{3 \log_2 M}{(M-1)(2M-1)} \frac{E_b}{n_0}} \right)$$



SEP Curves for MASK

- Increasing M deteriorates the performance, and for large M the distance between curves corresponding to M and $2M$ is roughly 6 dB.





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From real to complex

- For real baseband multi-symbol detection, the best estimate (detected symbol) is the **closest** symbol to the received symbol.

$$a^* = \arg \max_{a \in U} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-a)^2}{2\sigma^2}\right) = \arg \min_{a \in U} |y - a|$$

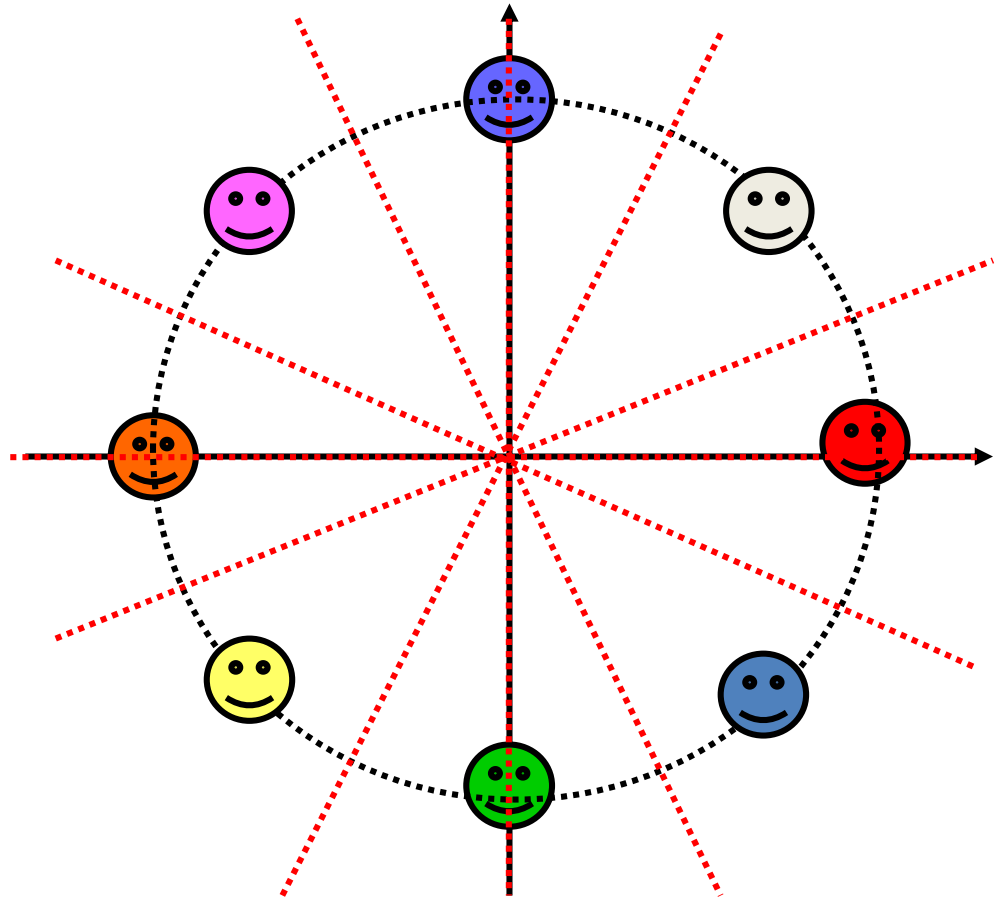
- For complex symbols, the rule is **the same!**



MPSK Error Analysis

- For MPSK, the detection threshold is the perpendicular bisector of segment between two adjacent points.

$$(a_n, b_n) = (A \cos \phi_n, A \sin \phi_n)$$
$$\phi_n \in \left\{ 0, \frac{2\pi}{M}, \frac{4\pi}{M}, \dots, \frac{2(M-1)\pi}{M} \right\}$$



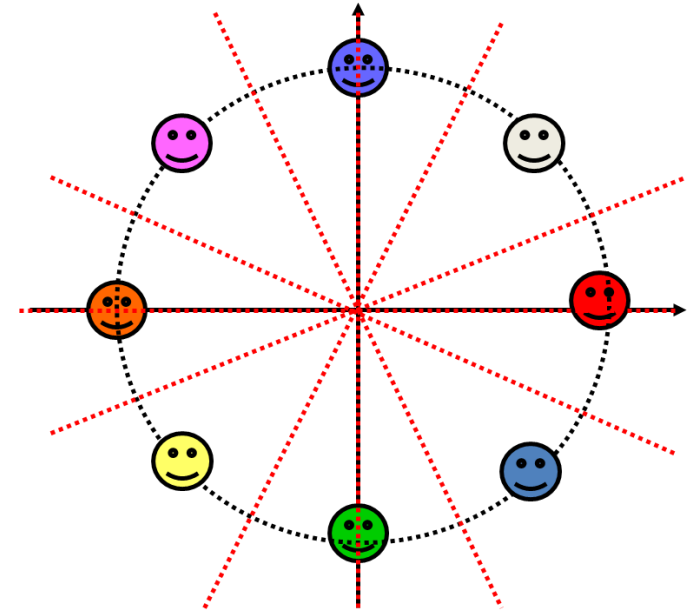


MPSK Error Analysis

■ Properties

- ▶ Symbols have unit power
- ▶ Symbols have same SEP
- ▶ Calculation is complicated, but once is enough!

■ Calculate SEP for $(A, 0)$



$$(a_n, b_n) = (A \cos \phi_n, A \sin \phi_n)$$
$$\phi_n \in \left\{ 0, \frac{2\pi}{M}, \frac{4\pi}{M}, \dots, \frac{2(M-1)\pi}{M} \right\}$$



*MPSK Error Analysis

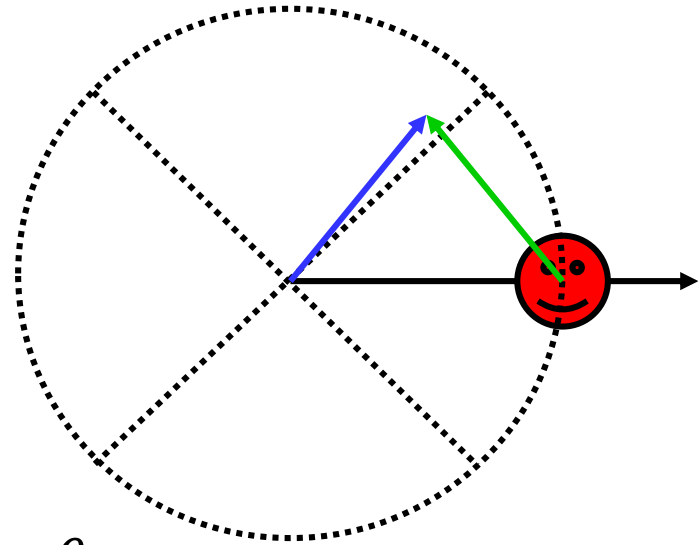
- Probability distribution of receive signal

$$f(y) = \frac{1}{\pi\sigma^2} \exp\left[-\frac{(y_I - A)^2 + y_Q^2}{\sigma^2}\right]$$

- Convert to polar coordinates

$$J(\rho, \theta) = \begin{vmatrix} \cos \theta & -\rho \sin \theta \\ \sin \theta & \rho \cos \theta \end{vmatrix} = \rho$$

- $$f(\rho, \theta) = \frac{\rho}{\pi\sigma^2} \exp\left[-\frac{\rho^2 + A^2 - 2A\rho \cos \theta}{\sigma^2}\right]$$



Law of Cosines: ρ, A, n



*MPSK Error Analysis

$$\begin{aligned} \blacksquare f(\rho, \theta) &= \frac{\rho}{\pi\sigma^2} \exp\left[-\frac{\rho^2 + A^2 - 2A\rho \cos \theta}{\sigma^2}\right] \\ &= \frac{\rho}{\pi\sigma^2} \exp\left[-\frac{(\rho - A \cos \theta)^2 + A^2 \sin^2 \theta}{\sigma^2}\right] \end{aligned}$$

■ Calculate marginal density for phase

$$\begin{aligned} f(\theta) &= \int_0^\infty \frac{\rho}{\pi\sigma^2} \exp\left[-\frac{(\rho - A \cos \theta)^2 + A^2 \sin^2 \theta}{\sigma^2}\right] d\rho \quad \text{Substitute } \sqrt{2}\rho/\sigma \text{ with } \rho \\ &= \exp\left(-\frac{A^2}{\sigma^2} \sin^2 \theta\right) \int_0^\infty \frac{\sqrt{2}\rho}{2\pi\sigma} \exp\left[-\frac{1}{2}\left(\frac{\sqrt{2}\rho}{\sigma} - \frac{\sqrt{2}A}{\sigma} \cos \theta\right)^2\right] d\left(\frac{\sqrt{2}\rho}{\sigma}\right) \\ &= \exp\left(-\frac{A^2}{\sigma^2} \sin^2 \theta\right) \frac{1}{2\pi} \int_0^\infty \rho \exp\left[-\frac{1}{2}\left(\rho - \frac{\sqrt{2}A}{\sigma} \cos \theta\right)^2\right] d\rho \end{aligned}$$

We make this look like a Gaussian distribution!



*MPSK Error Analysis

- Use E_s/n_0 to rewrite $f(\theta)$

$$f(\theta) = \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) \frac{1}{2\pi} \int_0^\infty \rho \exp\left[-\frac{1}{2}\left(\rho - \sqrt{\frac{2E_s}{n_0}} \cos \theta\right)^2\right] d\rho$$

- Approximation

- ▶ When $E_s/n_0 \rightarrow \infty$, Gaussian distribution tail $\rightarrow 0$

$$\begin{aligned} f(\theta) &\approx \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) \frac{1}{2\pi} \int_{-\infty}^\infty \rho \exp\left[-\frac{1}{2}\left(\rho - \sqrt{\frac{2E_s}{n_0}} \cos \theta\right)^2\right] d\rho \\ &= \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) \frac{1}{\sqrt{2\pi}} \sqrt{\frac{2E_s}{n_0}} \cos \theta \end{aligned}$$

This is the mean of Gaussian distribution.



*MPSK Error Analysis

■ Calculate SEP

$$\begin{aligned} P_{S,MPSK} &= 1 - \int_{-\pi/M}^{\pi/M} f(\theta) d\theta = 1 - \int_{-\pi/M}^{\pi/M} \frac{1}{\sqrt{2\pi}} \sqrt{\frac{2E_s}{n_0}} \cos \theta \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) d\theta \\ &\approx \frac{2}{\sqrt{\pi}} \int_{\pi/M}^{\infty} \sqrt{\frac{E_s}{n_0}} \cos \theta \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) d\theta && \text{Approximation when } M \text{ is large!} \\ &= \frac{2}{\sqrt{\pi}} \int_{\pi/M}^{\infty} \exp\left(-\frac{E_s}{n_0} \sin^2 \theta\right) d\left(\sqrt{\frac{E_s}{n_0}} \sin \theta\right) && \text{Variable substitution} \\ &= \frac{2}{\sqrt{\pi}} \int_{\sqrt{E_s/n_0} \sin \pi/M}^{\infty} \exp(-u^2) du \\ &= \operatorname{erfc}\left(\sqrt{\frac{E_s}{n_0}} \sin \frac{\pi}{M}\right) = 2Q\left(\sqrt{\frac{2E_s}{n_0}} \sin \frac{\pi}{M}\right) && \text{erfc definition and relationship with Q-function} \end{aligned}$$



*MPSK Error Analysis

- Express with E_b/N_0 and convert to BER

$$P_{b,\text{MPSK}} = \frac{2}{\log_2 M} Q \left(\sqrt{2 \log_2 M \frac{E_b}{n_0} \sin \frac{\pi}{M}} \right)$$

- Too complicated! Any simpler solution?



MPSK Error Analysis

■ A simple deduction with basic geometry!

▶ Decompose noise into

$$n = n_{\parallel} + n_{\perp}$$

▶ Error occurs when

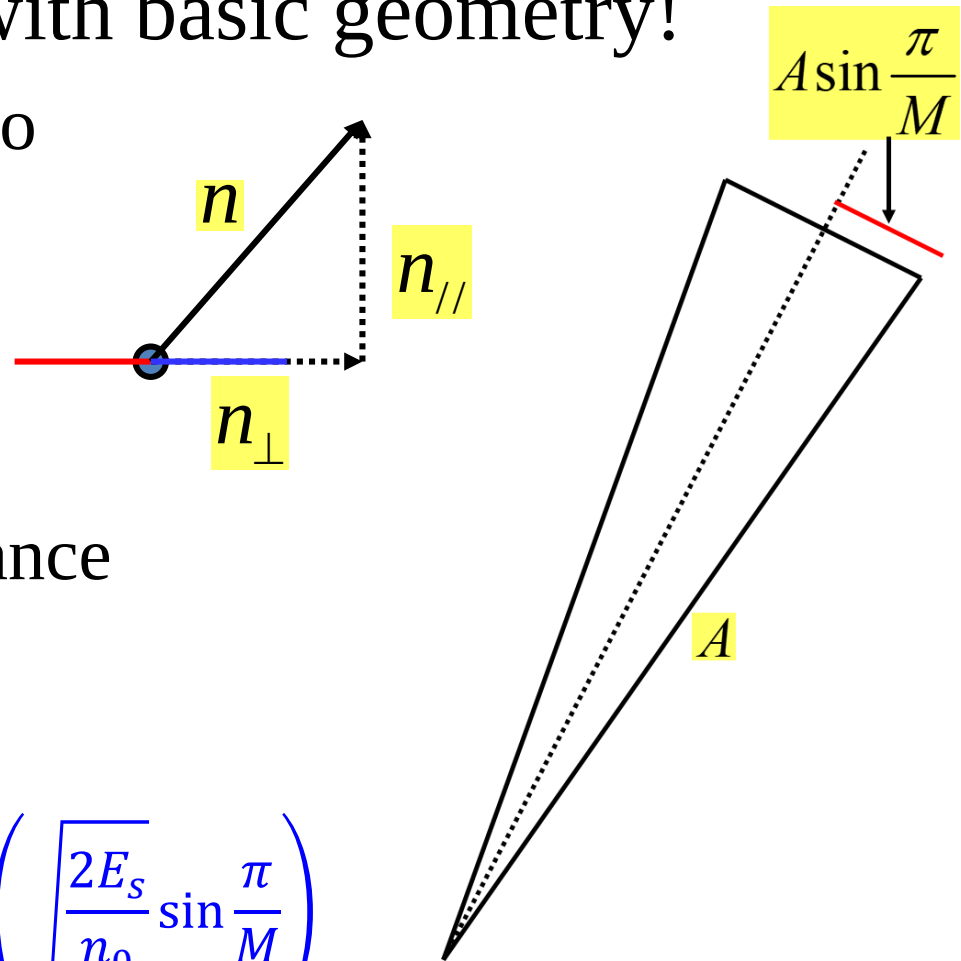
$$n_{\perp} > A \sin \frac{\pi}{M}$$

▶ The component variance

$$n_{\perp} \sim \mathcal{N}\left(0, \frac{n_0}{2}\right)$$

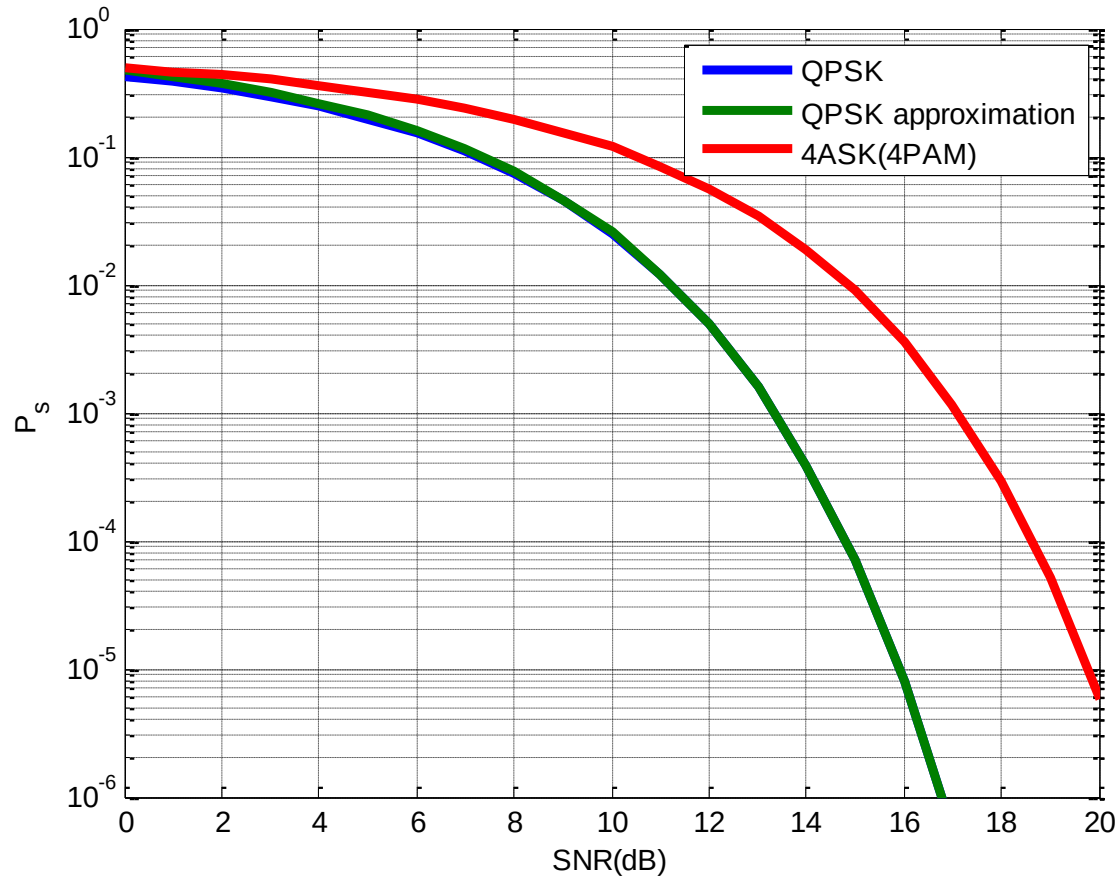
▶ Hence SEP

$$P_{S,MPSK} = 2Q\left(\frac{A}{\sigma_{\perp}} \sin \frac{\pi}{M}\right) = 2Q\left(\sqrt{\frac{2E_s}{n_0}} \sin \frac{\pi}{M}\right)$$





How Good is the Approximation?



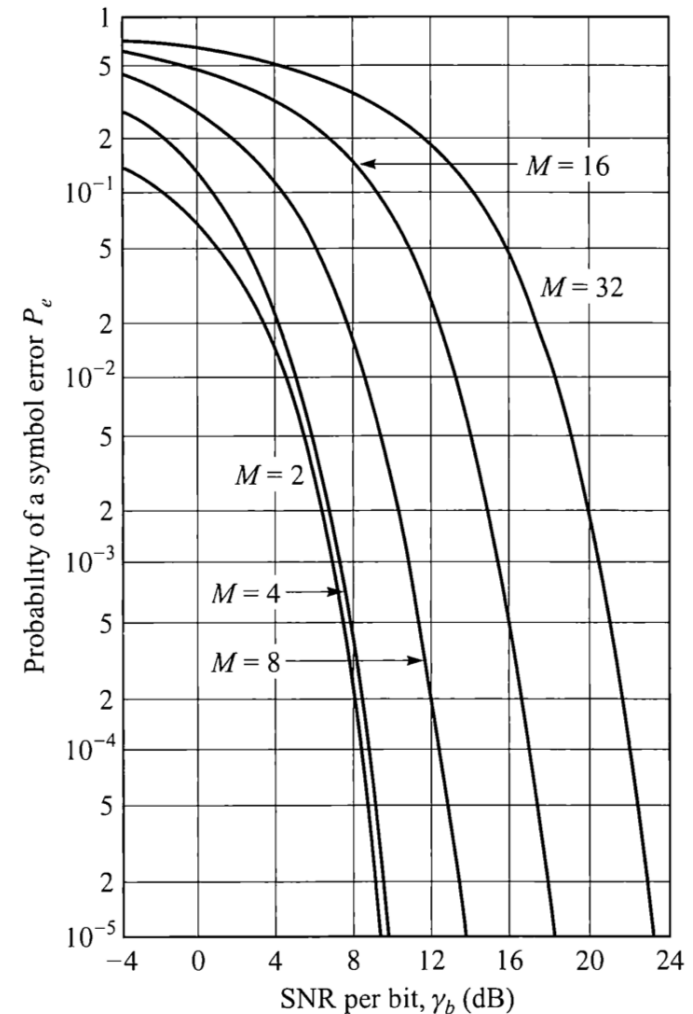
$$P_{s,4PSK} = 2Q\left(\sqrt{\frac{S}{2N}}\right)\left[1 - \frac{1}{2}Q\left(\sqrt{\frac{S}{2N}}\right)\right]$$

M=4 Precise SEP



SEP Curves

- Increasing M deteriorates the performance, and for large M the distance between curves corresponding to M and $2M$ is roughly 6 dB.
- Similar to ASK.





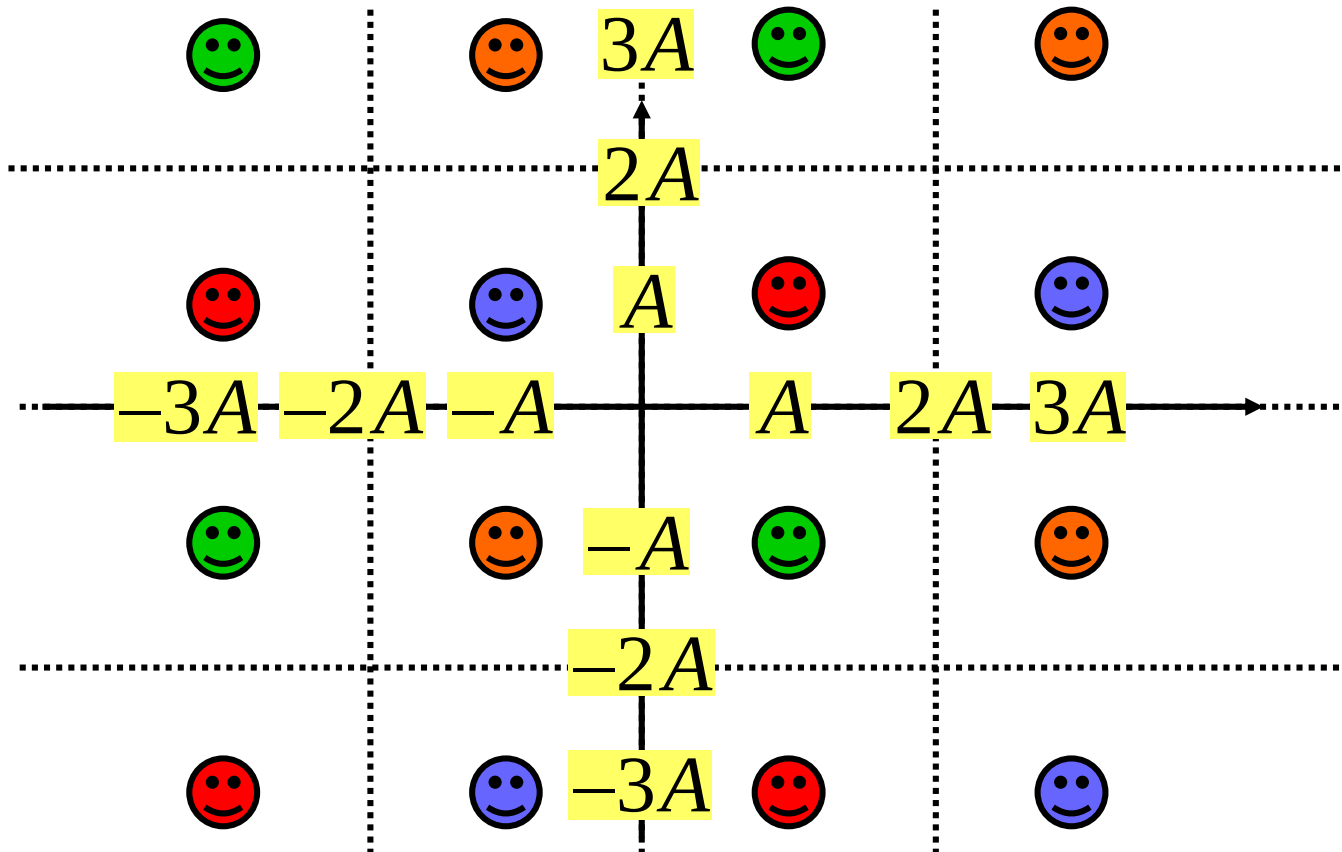
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MQAM Error Analysis

- For MQAM, the detection threshold is a grid.



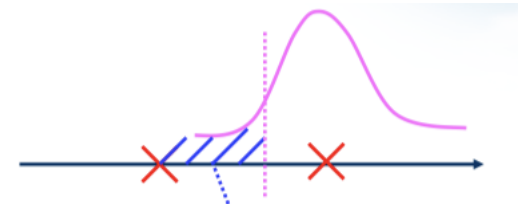


MQAM Error Analysis

■ Recall:

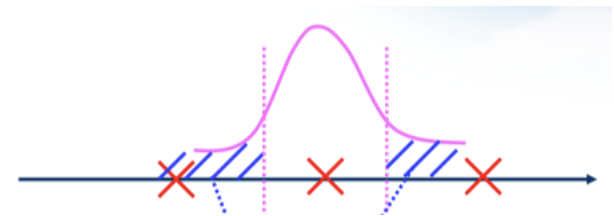
▶ Uni-lateral Error

$$Q\left(\frac{A}{\sigma_L}\right)$$



▶ Bilateral Error

$$2Q\left(\frac{A}{\sigma_L}\right)$$



■ Consider MQAM as two linear ASK

$$\sigma_L = \sigma_I = \sigma_Q = \sqrt{\frac{n_0}{2}} = \frac{\sigma}{\sqrt{2}}$$



MQAM Error Analysis

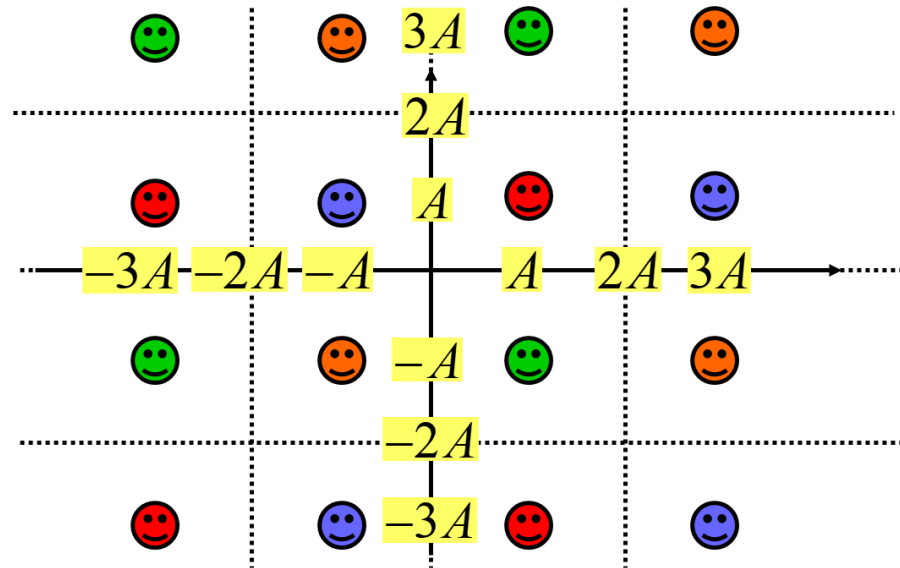
■ 4 points at the corner

- ▶ For each dimension, the probability of no error is

$$1 - Q\left(\frac{A}{\sigma_L}\right)$$

- ▶ The symbol error probability is

$$P_s = 1 - \left[1 - Q\left(\frac{A}{\sigma_L}\right)\right]^2$$





MQAM Error Analysis

■ $4(L - 2)$ points on 4 sides, $L = \sqrt{M}$

► For one dimension, the probability of no error is

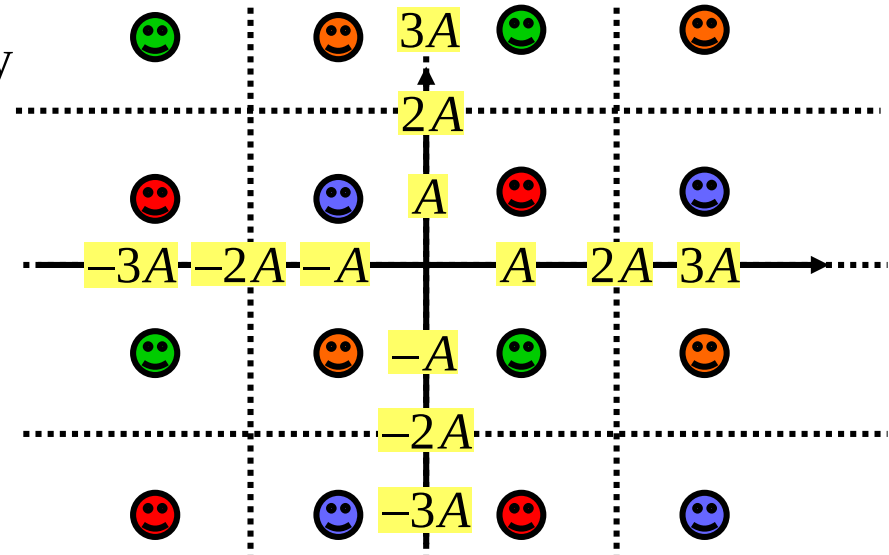
$$1 - Q\left(\frac{A}{\sigma_L}\right)$$

► For another dimension,

$$1 - 2Q\left(\frac{A}{\sigma_L}\right)$$

► Hence

$$P_s = 1 - \left[1 - Q\left(\frac{A}{\sigma_L}\right)\right] \left[1 - 2Q\left(\frac{A}{\sigma_L}\right)\right]$$





MQAM Error Analysis

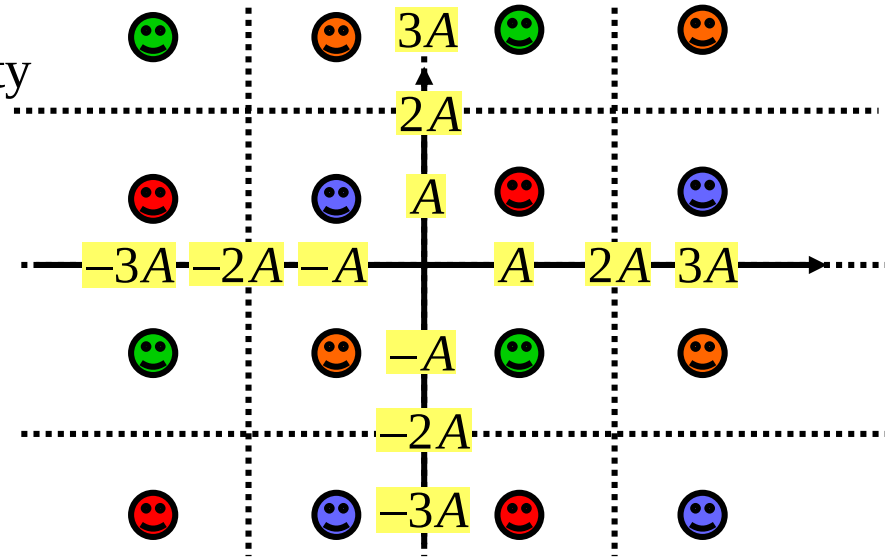
■ $(L - 2)^2$ internal points

- ▶ For each dimension, the probability of no error is

$$1 - 2Q\left(\frac{A}{\sigma_L}\right)$$

- ▶ The symbol error rate is

$$P_s = 1 - \left[1 - 2Q\left(\frac{A}{\sigma_L}\right)\right]^2$$





MQAM Error Analysis

■ Average SEP

$$\begin{aligned} P_s &= \frac{1}{L^2} [4(2Q - Q^2) + 4(L - 2)(3Q - 2Q^2) + (L - 2)^2(4Q - 4Q^2)] \\ &\approx \frac{1}{L^2} [8Q + 12(L - 2)Q + 4(L - 2)^2Q] \\ &= \frac{4L^2 - 4L}{L^2} Q \left(\frac{A}{\sigma_L} \right) \end{aligned}$$

■ Proakis Method

$$P_s = 1 - (1 - P_{s,LPAM})^2 = \frac{4L^2 - 4L}{L^2} Q \left(\frac{A}{\sigma_L} \right)$$



MQAM Error Analysis

■ Represent E_s with A

$$\begin{aligned} E_s &= 2 \times \frac{2A^2}{L} [1^2 + 3^2 + \dots + (L-1)^2] \\ &= 2 \times \frac{2A^2}{L} \frac{L}{6} (L^2 - 1) \\ &= \frac{2(L^2 - 1)}{3} A^2 \end{aligned}$$



MQAM Error Analysis

■ Express SEP with E_s/n_0

$$\frac{A}{\sigma_L} = \sqrt{\frac{3}{L^2 - 1} \frac{E_s}{n_0}}$$

$$P_{s,MQAM} = \frac{4L^2 - 4L}{L^2} Q \left(\sqrt{\frac{3}{L^2 - 1} \frac{E_s}{n_0}} \right)$$



MQAM Error Analysis

- Express SEP with E_b/N_0

$$P_{s,MQAM} = 4 \left(1 - \frac{1}{\sqrt{M}}\right) Q \left(\sqrt{\frac{3 \log_2 M E_b}{M-1} \frac{1}{n_0}} \right) \approx 4Q \left(\sqrt{\frac{3 \log_2 M E_b}{M-1} \frac{1}{n_0}} \right)$$

Large M approximation

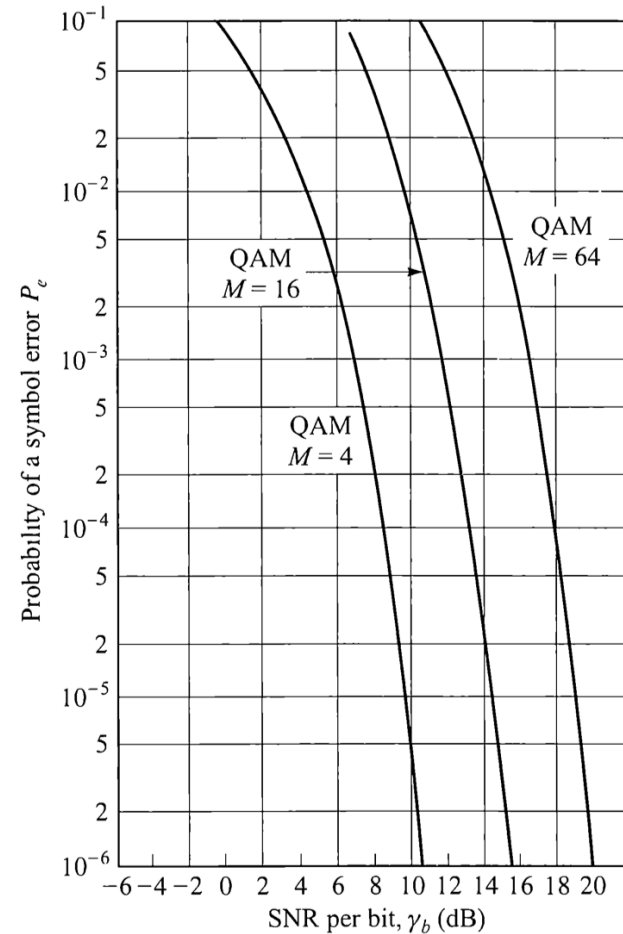
- Convert from SEP to BEP

$$P_{b,MQAM} = \frac{4}{\log_2 M} Q \left(\sqrt{\frac{3 \log_2 M E_b}{M-1} \frac{1}{n_0}} \right)$$



SEP Curves

- Unlike ASK and PSK, the penalty for doubling M is 3 dB!
- QAM is more power efficient.





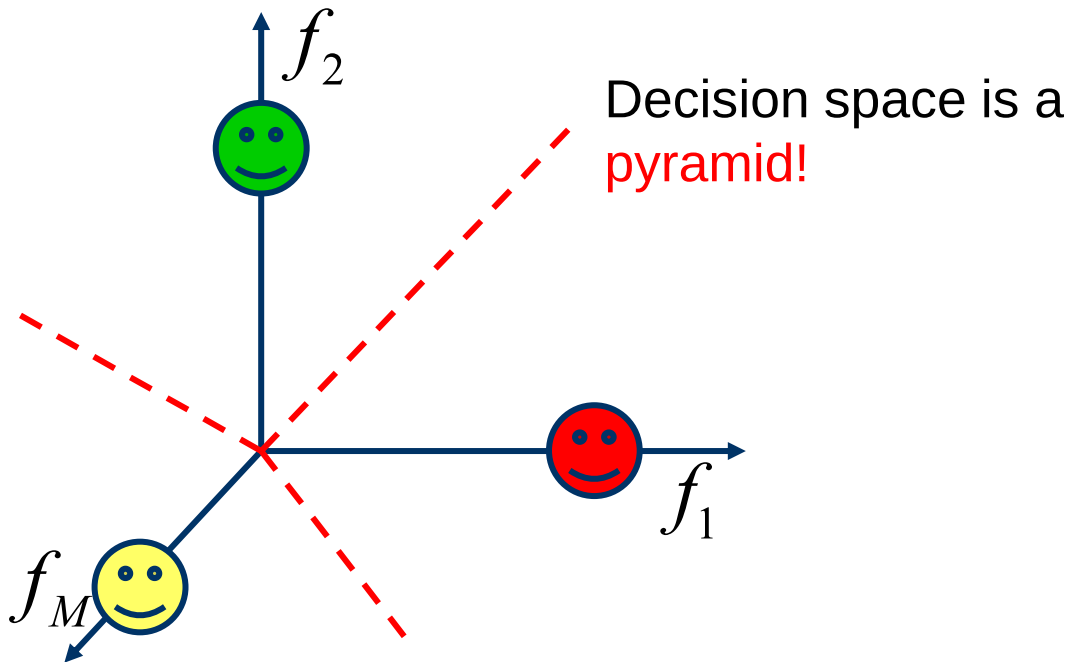
Outline

- Baseband equivalent AWGN channel
- Detection and error analysis
 - ▶ MASK
 - ▶ MPSK
 - ▶ MQAM
 - ▶ *MFSK



MFSK Error Analysis

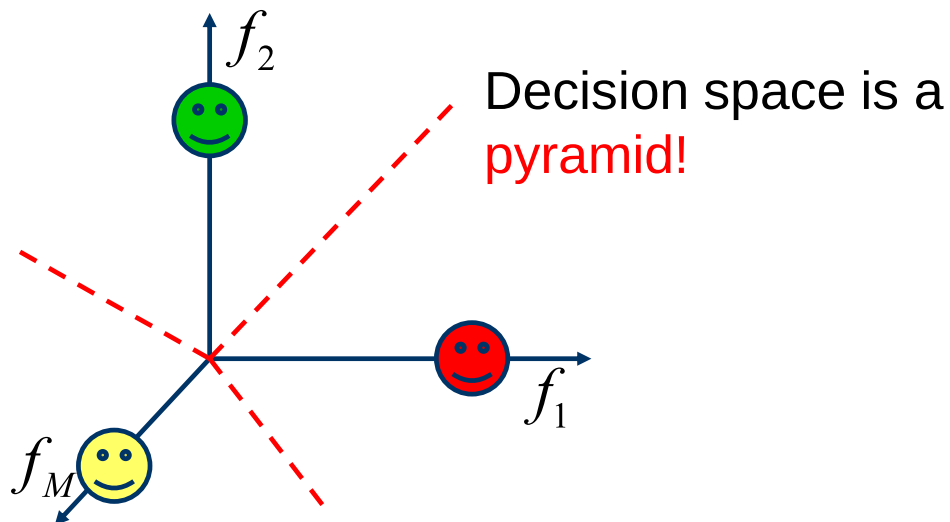
- Consider a signal space where MFSK signals are perpendicular vectors





*MFSK Error Analysis

- FSK vector $[A, 0, 0, \dots, 0]$
- Matched filter output $[A + n_1, n_2, n_3, \dots, n_M]$
- Correct decision: within the pyramid
$$A + n_1 > n_i, i = 2, 3, \dots, M$$





*MFSK Error Analysis

■ PDF of output signal

$$f(\mathbf{y}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_1 - A)^2}{2\sigma^2}\right) \prod_{i=2}^M \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y_i^2}{2\sigma^2}\right)$$

■ Error probability (not satisfying $A + n_1 > n_i, i = 2, 3, \dots, M$)

$$\begin{aligned} P_s &= 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_1 - A)^2}{2\sigma^2}\right) \times \prod_{i=2}^M \int_{-\infty}^{y_1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y_i^2}{2\sigma^2}\right) dy_i dy_1 \\ &= 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_1 - A)^2}{2\sigma^2}\right) \times \left[\int_{-\infty}^{y_1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{y_i^2}{2\sigma^2}\right) dy_i \right]^{M-1} dy_1 \end{aligned}$$



*MFSK Error Analysis

- Let $x = \frac{y_1}{\sigma}$, then rewrite SEP with Q-function

$$P_S = 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - A/\sigma)^2}{2}\right) \times [1 - Q(x)]^{M-1} dx$$

- With linear translation

$$P_S = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) \times \left[1 - Q\left(u + \frac{A}{\sigma}\right)\right]^{M-1} du$$



*MFSK Error Analysis

■ To express in E_s/n_0 and E_b/n_0 , use

$$\frac{A}{\sigma} = \sqrt{\frac{E_s}{n_0/2}} = \sqrt{\log_2 M \frac{2E_b}{n_0}}$$

■ Then SEP becomes

$$P_{s,MFSK} = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) \times \left[1 - Q\left(u + \sqrt{\frac{2E_s}{n_0}}\right)\right]^{M-1} du$$

$$P_{s,MFSK} = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) \times \left[1 - Q\left(u + \sqrt{\log_2 M \frac{2E_b}{n_0}}\right)\right]^{M-1} du$$



*SEP to BEP

- All “symbols” are adjacent, so

$$P_s(x_i \rightarrow x_j) = \frac{P_{s,MFSK}}{M-1}$$

- There are $\binom{\log_2 M}{n}$ symbols with n error bits.

$$\begin{aligned} P_{b,MFSK} &= \frac{1}{\log_2 M} \sum_{n=1}^{\log M} n \binom{\log_2 M}{n} \frac{P_{s,MFSK}}{M-1} \\ &= \frac{M}{2M-2} \left\{ 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) \times \left[1 - Q\left(u + \sqrt{\log_2 M \frac{2E_b}{n_0}}\right) \right]^{M-1} du \right\} \\ &= \frac{1}{2} - \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) \times \left[1 - Q\left(u + \sqrt{\log_2 M \frac{2E_b}{n_0}}\right) \right]^{M-1} du \end{aligned}$$



*MFSK Error Analysis

■ Comments

- ▶ For low SEP, M the larger the better!
- ▶ Trade bandwidth for SEP

■ Upper bound

$$P_s \leq \begin{cases} \exp \left[-\log M \left(\sqrt{\frac{E_s}{\log M n_0}} - \sqrt{\ln 2} \right)^2 \right] \\ \exp \left[-\left(\frac{E_s}{2n_0} - \log M \ln 2 \right) \right] \end{cases} \quad \begin{aligned} \frac{E_s}{4n_0 \log M} &\leq \ln 2 < \frac{E_s}{n_0 \log M} \\ \ln 2 &< \frac{E_s}{4n_0 \log M} \end{aligned}$$

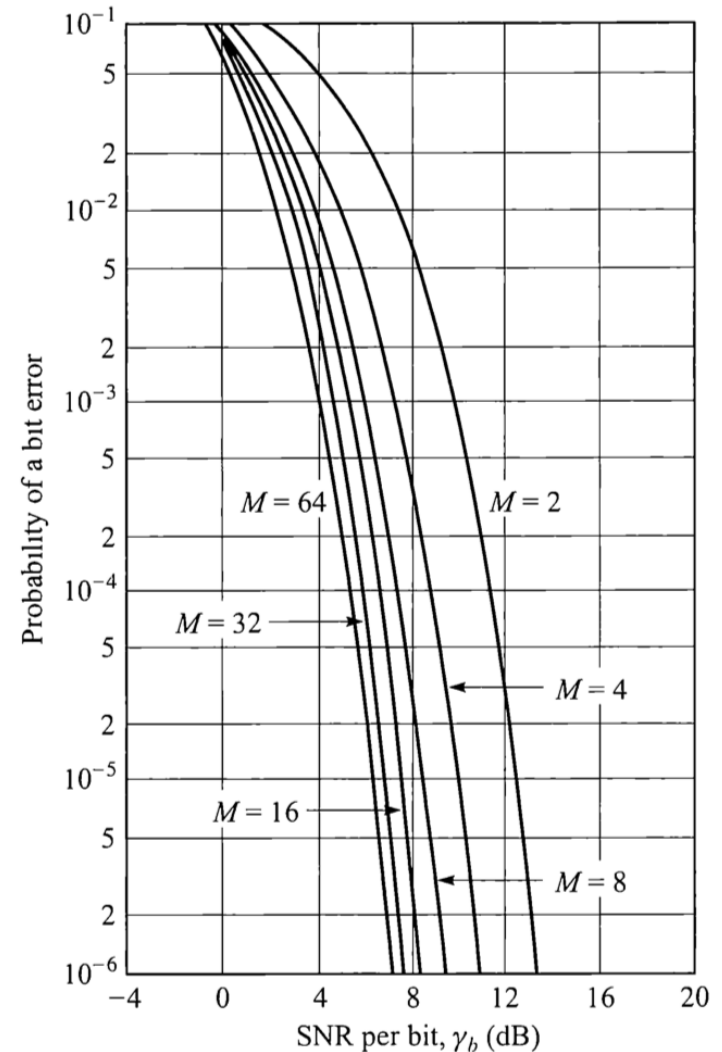


BER Curves

- Contrast to ASK, PSK, and QAM, increasing M **saves** transmitting energy to achieve certain BER (**lowers** the SNR requirement).

- Cost: bandwidth efficiency

$$\frac{R_b}{B} = \frac{\log_2 M}{M} \text{ bit/s/Hz}$$





Summary

- Baseband equivalent AWGN channel
 - ▶ Same as baseband AWGN but 2D noise
- Detection and error analysis ([Haykin: 6.3-6.5](#), [Proakis: 4.3-4.4](#))
 - ▶ MASK: antipodal and non-antipodal
 - ▶ MPSK: geometric approach
 - ▶ MQAM: 2D linear ASK
 - ▶ *MFSK: signal space analysis